AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer EIGHT questions in all. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER Q.1 OR Q.2)

Q.1. (Total 6 Marks)

a. Uneza and Junaid have Rs.150 and Rs.300 as their monthly savings respectively. Each of them spent a certain amount. Junaid then had 4 times as much remaining as Uneza. Find the amount spent by each of them.

(4 Marks)

b. Solve the following quadratic equation by the factorization method. (2 Marks)

$$x^2 - 8x + 16 = 0$$

(ATTEMPT EITHER Q.1 OR Q.2)

Q.2. (Total 6 Marks)

a. Find the solution set of the following simultaneous linear equations. (3 Marks)

$$y - 3 = 0$$
$$x - y + 1 = 0$$

b. A wire of length x metres is bent to form a square of perimeter x metres. If area of the square is twice its perimeter, then find the length of the wire. (3 Marks)

Q.3. (Total 4 Marks)

EITHER

a. Simplify $\begin{bmatrix} 2 & 3 \\ 9 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

OR

b. Find the multiplicative inverse of the matrix $\begin{bmatrix} 3a & a \\ 1 & 1 \end{bmatrix}$.

Q.4. (Total 4 Marks)

Eliminate a from the following system of equations and write the eliminant in simplified form.

$$x - a = 3$$

$$ax + \frac{1}{a} = 0$$

Q.5. (Total 5 Marks)

ETHIER

a. If $x-2$ is inversely proportional to $y+1$, then find the missing value of y in the following table.

x	3	5
y	2	?

OR

b. If $x-1, y$ and x^2+x+1 are in continued proportion, then find an expression of y in terms of x and also find y when $x=2$.

Q.6. (Total 5 Marks)

EITHER

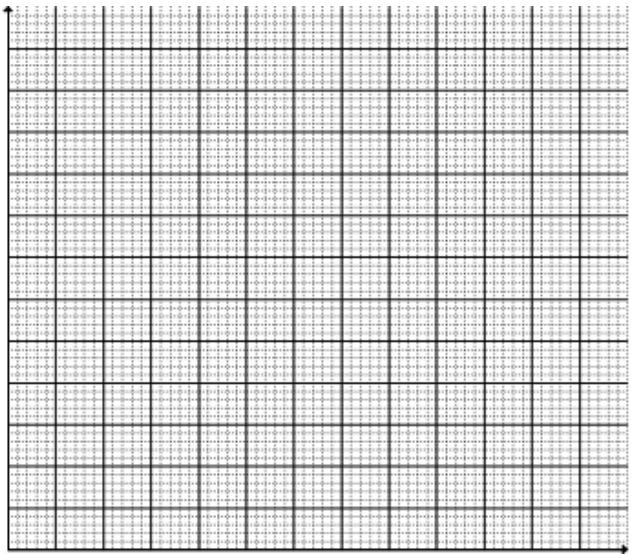
a. There are 7 squares of different lengths. If areas of the squares are $1cm^2$, $9\ cm^2$, $9\ cm^2$, $36\ cm^2$, $36\ cm^2$, $49\ cm^2$ and $64\ cm^2$, then find the variance in their lengths.

OR

b. On 14th August, 80 cakes were sold in a bakery. The masses of cakes are as follows:

Mass (Pounds)	1-3	3-5	5-7	7-9	9-11
Frequency	12	18	20	20	10

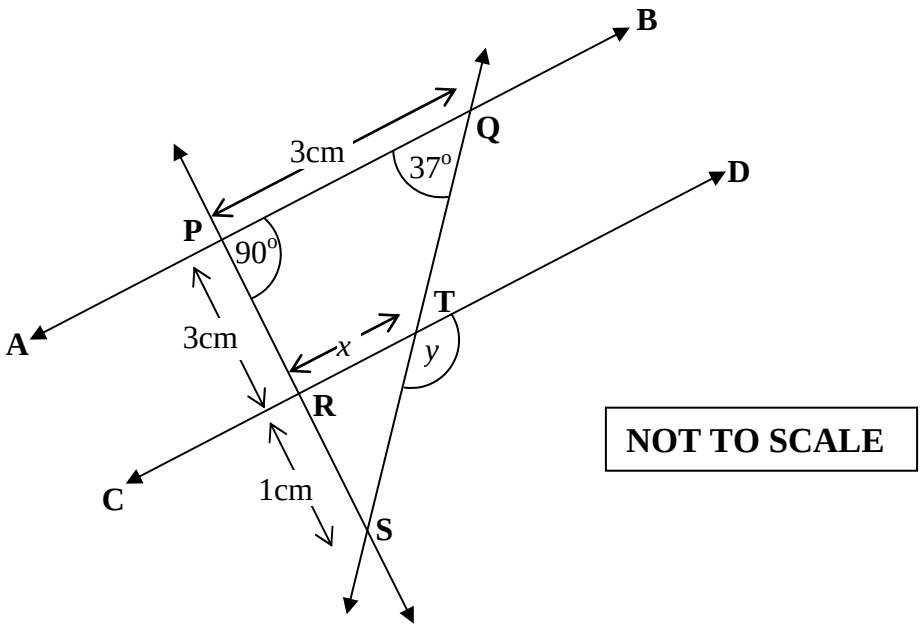
Write and use appropriate scale to draw a histogram on the graph below. Also draw a frequency polygon on the same graph.



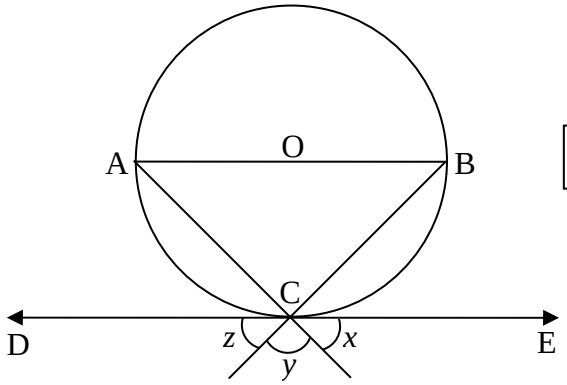
(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

Q.7. (Total 10 Marks)

a. In the given figure, AB and CD are parallel lines. Find the value of angle labelled as *y*. Also find the length of *x*. (5 Marks)

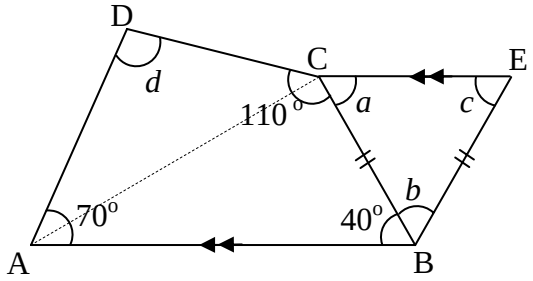


b. In the following figure A, B and C are points on the circumference of a circle with centre at O and $m\overline{AC} = m\overline{BC}$. If DE is parallel to AB, then find the value of *x*, *y* and *z*. Give reasons for your answer. (5 Marks)



NOT TO SCALE

c. In the following figure AB is parallel to CE. (5 Marks)



NOT TO SCALE

Find the measure of angles *a*, *b*, *c* and *d*. Give reason for your answer.

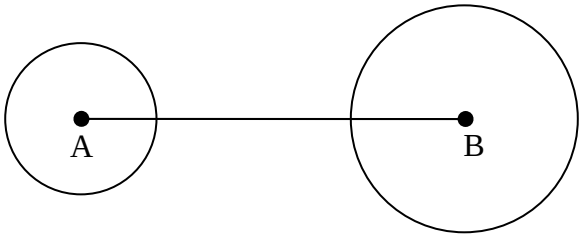
Q.8. (Total 5 Marks)

EITHER

a. In the following box draw a line segment AB of length 5.5cm. Draw a circle of radius 2.5cm with centre A and then draw two tangents to the circle from the point B.

OR

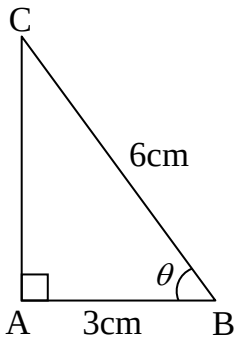
b. In the following figure, A and B are the centres of the circles with radii 1cm and 1.5cm respectively. Draw transverse common tangents to the circles.



Q.9. (Total 6 Marks)

EITHER

a. For the following right angled triangle ABC, find the value of $\sin^2\theta - \cos^2\theta$.



OR

b. In a right angled triangle if $\tan x = 1$, then find the value of x and prove that $\sin x + \cos x = \frac{2}{\sqrt{2}}$.

Please use this page for rough work